Weblinks

- https://www.youtube.com/watch?v=Ym6nlwvQZnE
- https://www.youtube.com/watch?v=2PSjARmmL7M

Do You Know!

"G" is also known as Newtonian constant of gravitation or the Cavendish gravitational constant.

Key points

Gravitational force has following characteristics:

i) It is always present between every two objects because of their masses.
ii) It exists everywhere in the universe.
iii) It forms an action-reaction pair.
iv) It is independent of the medium between the objects.
v) It is directly proportional to the product of the masses of objects.
vi) It is inversely proportional to the square of the distance between the centres of the objects.
vii) Hence it follows the "Inverse Square Law".

Worked Example 1

Determine the gravitational force of attraction between two spherical bodies of masses 500kg and 800kg. Distance between their centers is 2 meters.

Solution

Step:1 Write down known quantities and quantities to be found.

$m_1 = 500\text{kg}$
$m_2 = 800\text{kg}$
$r = 2\text{m}$
$G = 6.673 \times 10^{-11}\text{Nm}^2\text{kg}^{-2}$
$F = ??$

Step:2 Write down formula and rearrange if necessary

$$F = G\frac{m_1m_2}{r^2}$$

Step:3 Put the values in formula and calculate

$$F = \frac{6.673 \times 10^{-11}\text{Nm}^2\text{kg}^{-2} \times 500\text{kg} \times 800\text{kg}}{(2\text{m})^2}$$

$$F = 6.67 \times 10^{-6}\text{N}$$

Hence, the gravitational force of attraction between the bodies is 6.67×10^{-6}N.

Law of Gravitation and Newton's Third law of Motion

According to Newton's law of gravitation, every two objects attract each other with equal force but in opposite direction. As shown in fig (6.2).

From the figure:

$m_1 \rightarrow$ Mass of body (A)

$m_2 \rightarrow$ Mass of body (B)

$F_{12} \rightarrow$ Force with which body (A) attracts body (B)

$F_{21} \rightarrow$ Force with which body (B) attracts body (A)

Then according to this law

$$F_{12} = -F_{21}$$

This shows that, the two forces are equal in magnitude but opposite in direction. Now, if F_{12} is considered as "Action Force" and F_{12} as "Reaction Force". Then by using above equation, it is concluded that "Action equals to reaction but in opposite direction".

Recall that, above statement is in accordance with the Newton's third law of motion which states that "To every action there is always an equal and opposite reaction".

Hence, Newton's law of gravitation is consistent with Newton's third law of motion.

For example, according to Newton's law of Universal gravitation the Earth pulls the Moon with its gravity and the Moon pulls the Earth with its gravity. Therefore they form an action-reaction pair, which is in accordance with Newton's third law of motion.

Do You Know!

Henry Cavendish in 1798 completed the 1st experiment that demonstrated the Newton's Law of universal gravitation. This happens more than a century after Newton had announced law of universal gravitation.

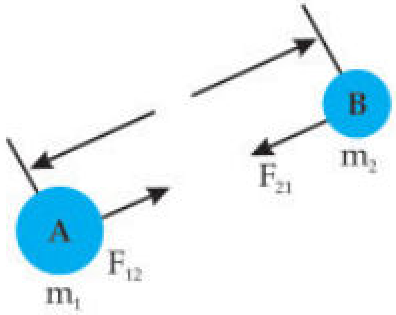


Fig (6.2)

Demonstration of Newton's law of universal gravitation in accordance with Newton's third law of motion.

Gravitational Field

Gravitational field can be described as:

"A gravitational field is a region in which a mass experiences a force due to gravitational attraction".

The earth has an attractive gravitational field around it. Any object near the Earth experience this

Activity

Measure the masses of your copy and pen and then calculate the gravitational force of attraction between them.

force which is due to Earth's gravity. This field is directed towards the centre of the Earth as shown in fig (6.3).

This field is strongest near the surface of the Earth and gets weaker as we move farther and farther away from the Earth. This force is called the "Field Force" because it acts on all objects whether they are in contact with Earth's surface or not. So, it is a non-contact force. For example, it acts on an aeroplane either it is standing on Earth's surface or flying in the sky.

A body of mass one kilogram (1kg) on Earth experiences a force of about ten newton (10N) due to Earth's gravitational field. This force determines the gravitational field strength which is defined as:

> Gravitational field strength 'g' is the gravitational force acting per unit mass.

The gravitational field strength "g" is approximately 10 Newton per kilogram ($10Nkg^{-1}$).

The gravitational field strength "g" is different at different planets. For example, the gravitational field strength "g" on the surface of Moon is approximately 1.6 Newton per Kilogram (Nkg^{-1}). Acceleration due to gravity "g" at different planets is shown in the table (6.1).

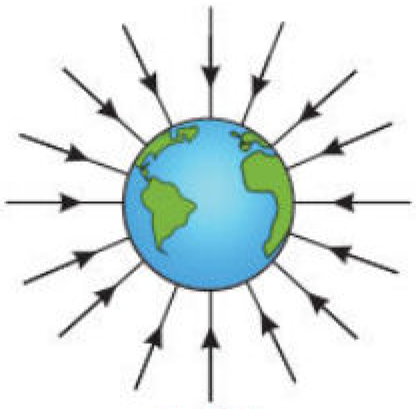


Fig (6.3)
Gravitational field around the Earth is directed toward center of earth from all direction

Point to Ponder

- Does the whole solar system works in a push and pull network?

Solar System	Value of g ms^{-2}
Earth	10
Moon	1.62
Venus	8.87
Mars	3.77
Jupiter	25.95
Sun	274
Mercury	3.59
Saturn	11.08
Uranus	10.67
Neptune	14.07

Table (6.1)
Acceleration due to gravity "g" at different planets.

Self Assessment Questions:

Q1: What will be the effect on gravitational pull between two objects if medium between them is changed?

Q2: Which force causes the moon to move in orbit around earth?

6.2 WEIGHT

We know that all the objects which are thrown upward in the air, fall back to the ground. Have you noticed why this is so?

The force applied by the Earth's gravitational field, pulls the objects downward. Weight is another name for the Earth's gravitational force on the objects. Therefore weight can be defined as:

The weight of an object is the measurement of gravitational force acting on the object.

Weight 'W' of an object of mass 'm', in a gravitational field of strength 'g' is given by the relation:

$$W = mg \quad\text{.......................}(6.2)$$

Like other forces weight is also measured in Newton's (N). Spring balance is used to measure weight of an object; Fig 6.4.

An object of mass 1kg has a weight of 9.8N near the surface of Earth. Though 10N is accurate enough for many calculations. Therefore we use 10N in this book. The objects with larger masses may have larger weights. Your weight vary slightly from place to place, because Earth's gravitational field strength varies at different places. The weight of the object changes as it moves away from the Earth. The weight of the object is different at different planets. For example: you will have less weight at Moon because Moon's gravitational field is weaker than Earth.

Fig (6.4)
Spring balance

Do You Know!

'Gravity' is taken from Latin word 'gravitas' means 'weight'.

Worked Example 2

Calculate the weight of Rumaisa, who has a mass of 65kg standing at the ground? The strength of gravitational field on Rumaisa is 10 Newton per kilogram.

Solution

Step:1 Write down known quantities and quantities to be found.

$m = 65\text{kg}$

$g = 10\text{Nkg}^{-1}$

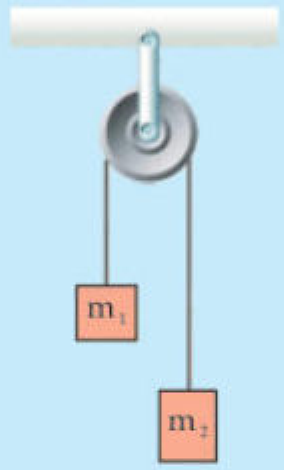
$W = ??$

Activity

The teacher should encourage and facilitate the students in the class to measure their masses and then calculate their weights on Earth, Moon and Mars.

Do You Know?

British scientist George Atwood (1746-1807) used two masses suspended from a fixed pulley, to study the motion and measure the value of 'g'. This is named as "Atwood Machine".



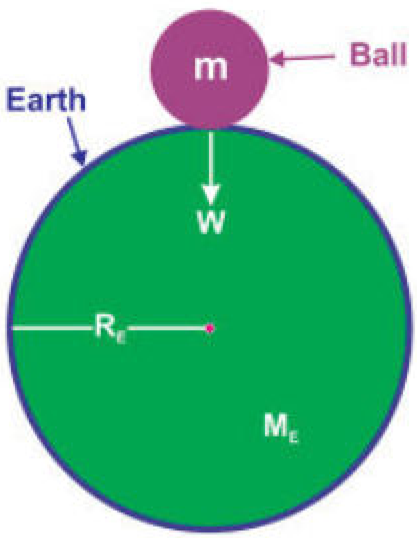


Fig (6.5)
Weight of a ball is equal to the gravitational force between the ball and the Earth.

Step:2 Write down formula and rearrange if necessary

$$W = mg$$

Step:3 Put the values in formula and calculate

$$W = 65kg \times 10Nkg^{-1}$$

Hence, the weight of Rumaisa is 650 Newton.

Self Assessment Questions:

Q3: Why weight of an object is different at different planets?

Q4: What is the actual value of 'g' near the surface of Earth?

Q5: The strength of gravity on the Moon is 1.6 Nkg^{-1}. If an astronaut's mass is 80kg on Earth, what would it be on the Moon?

Q6: If you go on a diet and lose weight, will you also lose mass? Explain.

6.3 Mass of Earth

Mass of Earth can not be measured directly by placing it on any weighing scale. But it can be measured by an indirect method. This method utilizes the Newton's law of universal gravitation. Let us consider; Fig (6.5), in which a small ball is placed on the surface of Earth.

M_b → Mass of the ball.
M_E → Mass of Earth.
G → Universal gravitational constant.
R_E → Radius of Earth; which is also the distance between the ball and centre of Earth.

Then according to Newton's law of universal gravitation, the gravitational force (F) of the Earth acts on the ball is:

$$F = G\frac{mM_E}{R_E^2} \quad ....(6.3)$$

Whereas the force with which Earth attracts the ball towards its centre is equal to the weight of the ball. Therefore

$$F = W = mg \text{ ....(6.4)}$$

Comparing equation (6.3) and (6.4); we get:

$$mg = G\frac{mM_E}{R_E^2}$$

Re-arrangment of above equation gives

$$M_E = \frac{gR_E^2}{G} \text{ ....(6.5)}$$

Numerical values of the constants at right hand side of equation (6.5) are:

$g = 10Nkg^{-1}$
$R_E = 6.38 \times 10^6 m$
$G = 6.673 \times 10^{-11} Nm^2kg^{-2}$

Substituting these values in equation (6.5), we get:

$$M_E = \frac{10Nkg^{-1} \times (6.38 \times 10^6 m)^2}{6.673 \times 10^{-11} Nm^2kg^{-2}}$$

$$M_E = 6.0 \times 10^{24} kg.$$

Thus, mass of Earth is 6.0×10^{24}kg.

Worked Example 3

Calculate the acceleration due to gravity on a planet that has mass two times to the mass of Earth and radius 1.5 times to the radius of Earth. If the acceleration due to gravity on the surface of Earth is $10ms^{-2}$. Calculate acceleration due to gravity on the planet.

Solution

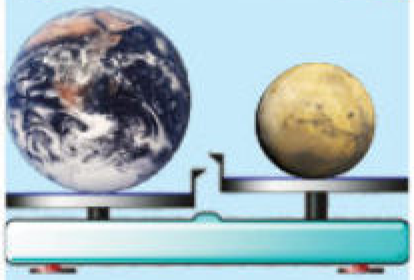
Step 1: Write down known quantities and quantities to be found.

Mass of the planet = $M_p = 2M_E$
Mass of the Earth = $M_E = 6.0 \times 10^{24}$kg

Do You Know!

Ocean tides are caused by the gravity of Moon.

Do You Know!



The Earth has (9.3) times more mass than Mars

Activity

Calculate the mass of the Earth if acceleration due to gravity; $g = 9.8ms^{-2}$.

Radius of the planet = $R_p = 1.5\ R_E$

Radius of the Earth = $R_E = 6.38 \times 10^6 m$

Universal gravitational constant

$= G = 6.673 \times 10^{-11} Nm^2kg^{-2}$

Acceleration due to gravity on the Earth = $g_E = 10ms^{-2}$

Acceleration due to gravity on the planet = g_p = ??

Step 2: Write down formula and rearrange if necessary

$$g_E = \frac{GM_E}{R_E^2}$$

For the planet:

$$g_p = \frac{GM_p}{R_p}$$

Step 3: Put the values in formula and calculate:

$$g_p = \frac{6.673 \times 10^{-11} Nm^2kg^{-2} \times 2 \times 6.0 \times 10^{24} kg}{(1.5 \times 6.38 \times 10^6 m)^2}$$

$$g_p = 8.74ms^{-2}$$

Hence, acceleration due to gravity on the planet is $8.74ms^{-2}$.

Self Assessment Questions:

Q7: What will be the value of acceleration due to gravity on the surface of earth if its radius reduces to half?

Q8: What will be acceleration due to gravity on the surface of earth f its mass reduces by 25%?

Q9: What will be the mass of a planet whose radius is 20% of the radius of earth?

Mass and Radius of different planets.

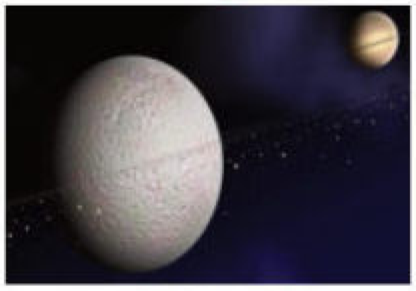


Fig (6.6) (a)
A natural satellite Moon revolving around the Earth

Fig (6.6) (b)
An artificial satellite revolving around the Earth

6.4 Artificial Satellites

A satellite is an object that revolves around a planet. Satellites are of two types:

1. Natural satellites Fig (6.6) (a).
2. Artificial satellites Fig (6.6) (b).

Natural Satellites	Artificial Satellites
The object which revolves around the planet naturally is called "Natural Satellite".	The object which are sent into space by scientists to revolve around the Earth or other planets are called "Artificial Satellite".
E.g Moon is a natural satellite because it revolves around the Earth naturally.	E.g. Sputnik-1, Explorer-1 are amongst the artificial satellites.

Artificial satellites are used for different purposes like

- For communication.
- For making star maps.
- For making maps of planetary surfaces.
- For collecting information about weather.
- For taking pictures of planets, etc.

Artificial satellites carry instruments, passengers or both to perform different experiments in space.

Artificial satellites have been launched into different orbits around the Earth. There are different types of orbits like:

- For communication.
- Low- Earth orbit.
- Medium- Earth orbit.
- Geostationary orbit.
- Elliptical orbit.

These orbits are characterized on the basis of different parameters like, their distance from the Earth, their time period around the Earth etc.

An artificial satellite which completes its one revolution around the Earth in 24 hours is used for communication purpose. As Earth also completes its one rotation about its axis in 24 hours, therefore the above satellite appears to be stationary with respect to

Interesting Information

Mass and radius of different objects

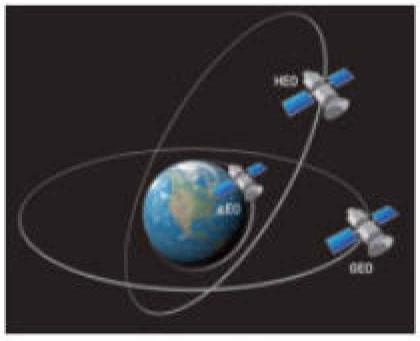
Sun planet and moon of our solar system	Mass (Kg)	Radius (m)
Sun	1.99×10^{30}	6.96×10^{8}
Moon	7.35×10^{22}	1.74×10^{6}
Mercury	3.30×10^{23}	2.44×10^{6}
Venus	4.87×10^{24}	6.05×10^{6}
Earth	5.97×10^{24}	6.38×10^{6}
Mars	6.42×10^{23}	3.40×10^{6}
Jupiter	1.90×10^{27}	6.91×10^{7}
Saturn	5.68×10^{26}	6.03×10^{7}
Uranus	8.68×10^{25}	2.56×10^{7}
Neptune	1.02×10^{26}	2.48×10^{7}

Do You Know!

First artificial satellite was Sputnik-1 which was sent into space by Soviet Union (Russia) on 4th October 1957.

Earth. Its orbit is therefore called "Geostationary orbit". As it is used for communication purpose, therefore it is known as "Communication Satellite".

Newton's Law of Gravitation in the motion of satellite



(Fig 6.7)
Motion of satellites in different orbits

The curved path along which a natural or artificial satellite revolves around a planet is called an "orbit"; (Fig 6.7). Rockets are used to put satellites into orbits in space. The Newton's law of gravitation has an important role in the motion of satellite in its orbit, because the gravitational pull of Earth on the satellite provides the centripetal force needed to keep a satellite in orbit around some planet.

Let us consider the motion of a satellite which is revolving around the Earth; Fig (6.8). In the figure:

m $\rightarrow$ Mass of the satellite.
M $\rightarrow$ Mass of Earth.
R $\rightarrow$ Radius of Earth
h $\rightarrow$ Height(altitude) of satellite from the surface of Earth.
r = R + h $\rightarrow$ Radius of orbit.

Then, as we already discussed:

Centripetal force = Gravitational force

$$\text{or} \quad F_C = F_G \longrightarrow (i)$$

$$\because F_C = \frac{mv^2}{r} \quad \text{and}$$

$$F_G = \frac{GmM}{r^2}$$

Substituting the values of (F_C) and (F_G)in equation (i):

$$\frac{\cancel{m}v^2}{\cancel{r}} = \frac{G\cancel{m}M}{\cancel{r^2}}$$

$$v^2 = \frac{GM}{r} \quad [\because r = R + h]$$

$$\therefore \quad v^2 = \frac{GM}{R+h}$$

Do You Know!

The height of a geostationary satellite is about 42,300 km from the surface of the Earth. Its velocity with respect to Earth is zero.

$$v = \sqrt{\frac{GM}{R+h}} \quad ....(6.6)$$

This gives the velocity that a satellite must possess when orbiting around Earth in an orbit of radius ($r = R+h$).

This shows that, the speed of the satellite is independent of its mass. Hence every satellite whether it is very massive (large) or very light (small) has the same speed in the same orbit.

The time required for a satellite to complete one revolution around the Earth in its orbit is called its time period "T". The time period of a satellite can be calculated as:

$$\because \quad T = \frac{2\pi r}{v} \text{ - - - - - - - - - - } (i)$$

The velocity of satellite is given by equation (6.6) as:

$$v = \sqrt{\frac{GM}{R+h}}$$

Substituting it in equation (i)

$$\therefore \quad T = \frac{2\pi r}{\sqrt{\frac{GM}{R+h}}} \qquad \because \; r = R+h$$

$$\therefore \quad T = 2\pi r\sqrt{\frac{r}{GM}}$$

$$\therefore \quad T = 2\pi\sqrt{\frac{r^3}{GM}} \quad ....(6.7)$$

Equations (6.7) gives the expression for the time period of a satellite orbiting around the Earth. Thus, Newton's law of gravitation helps to describe the motion of a satellite in an orbit around the Earth.

Motion of Artificial Satellite around the Earth

The satellites are put into their orbits around the Earth by rockets. When a satellite is put into orbit, its speed is selected carefully and correctly. If speed is not

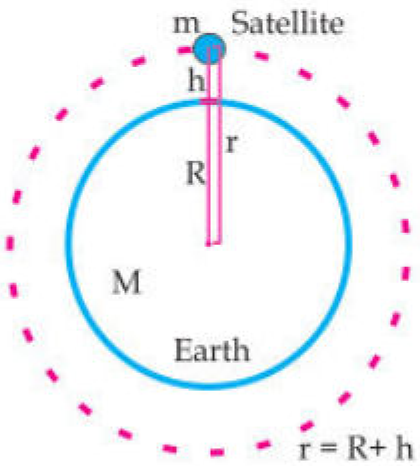


Fig (6.8)
A Satellite is orbiting around the Earth

Weblinks

How satellite is launched into an orbit:
https://www.youtube.com/watch?v=8t2eyEDy7p4

chosen correctly then the satellite may fall back to Earth or its path may take it further into orbit. During the motion of a satellite in the orbit the gravitation pull of Earth on it is always directed towards the centre of Earth.

Newton used the following example to explain how gravity makes the orbiting possible.

Imagine a cannonball launched from a high mountain; Figure 6.8; shows three paths the ball can follow.

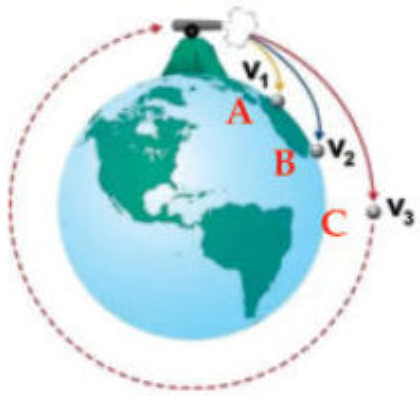


Fig 6.8
A cannon ball launched from high mountain

Path (A)	Path (B)	Path ©
The cannon ball is launched at a slow speed.	The cannon ball is launched at a medium speed.	The cannon ball is launched at a high speed.
The cannon ball will fall back to Earth.	The cannon ball will fall back to Earth.	The cannon ball will not fall back to Earth instead it orbits around the Earth.

Above example shows that, for an artificial satellite to orbit the Earth and to retrace its path it requires certain orbital velocity. The orbital velocity is defined as:

The velocity required to keep the satellite into its orbit is called "Orbital Velocity"

Do You Know!

Tides in ocean result from the gravitational attraction of sun and moon. Sun's gravitational attraction to the Earth is 177 times greater that of the moon to the Earth

The gravitational pull of Earth on a satellite provides the necessary centripetal force for orbital motion. Since this force is equal to the weight of satellite, '$W_S = mg$', therefore

$$F_C = W_S \text{ ------------ (i)}$$

and, $W_S = mg_h$

where,

m → Mass of the satellite.

g_h → Acceleration due to gravity at height 'h' from the surface of Earth.

The centripetal force 'F_C' on the satellite is:

$$F_C = \frac{mv^2}{r}$$

Substituting the values of 'F_C' and 'W_S' in equation (i):

$$\frac{mv^2}{r} = mg_h$$

$$\therefore \quad v^2 = g_h r$$

$$\therefore \quad v = \sqrt{g_h r}$$

$$\because \quad r = R + h$$

$$\therefore \quad \mathbf{v = \sqrt{g_h(R + h)}} \quad \ldots\ldots\ldots\ldots\ldots(6.8)$$

If satellite is orbiting very close to the surface of Earth then: $h << R$

In this case orbital radius may be considered equal to radius of Earth.

Therefore, $R + h = R$

Also $g_h = g$

and $v = v_C$

Where,

$v_C \rightarrow$ Critical velocity

$g \rightarrow$ Acceleration due to gravity on the surface of Earth.

In terms of above factors equation (6.8) becomes:

$$\mathbf{v_C = \sqrt{gR}} \quad \ldots\ldots\ldots\ldots\ldots(6.9)$$

This is known as "Critical velocity". It is defined as:

The constant horizontal velocity required to put the satellite into a stable circular orbit around the Earth.

It is also known as orbital speed or proper speed.

If

$$g = 10ms^{-2}$$

$$R = 6.38 \times 10^6 m$$

Then equation (6.9) becomes.

$$v_c = \sqrt{gR} = \sqrt{10ms^{-2} \times 6.38 \times 10^6 m}$$
$$v_c = 7.99 \times 10^3 ms^{-1}$$
$$\text{or} \quad v_c = 8.0kms^{-1}$$

It should be noted that as the satellite get closer to the Earth, the gravitational pull of the Earth on it gets stronger.

So, the satellites in order to stay in an orbit closer to Earth needs to travel faster as compare to those satellites in the farther orbits.

Worked Example 4

Calculate the speed of a satellite which orbits the Earth at an altitude of 1000 kilometres above Earth's surface?

Solution

Step: 1 Write down known quantities and quantities to be found:

$$M_{Earth} = M = 6.0 \times 10^{24} kg$$
$$R_{Earth} = R = 6.38 \times 10^6 m$$
$$h = 1000km = 1000 \times 10^3 m = 1 \times 10^6 m$$
$$G = 6.673 \times 10^{-11} Nm^2kg^{-2}$$
$$v = ??$$

Step:2 Write down formula and rearrange if necessary:

$$v = \sqrt{\frac{GM}{R+h}}$$

Step:3 Put the values in formula and calculate:

$$v = \sqrt{\frac{6.673 \times 10^{-11} Nm^2kg^{-2} \times 6.0 \times 10^{24} kg}{6.38 \times 10^6 m + 1 \times 10^6 m}}$$
$$v = 7.36 \times 10^3 ms^{-1}$$

Hence, the orbital speed of satellite is $7.36 \times 10^3 ms^{-1}$.

Self Assessment Questions:

Q10: Write down any four uses of artificial satellites.

Q11: What is Geostationary orbit?

Q12: Why the two satellites of different masses have same speed in the same orbit?

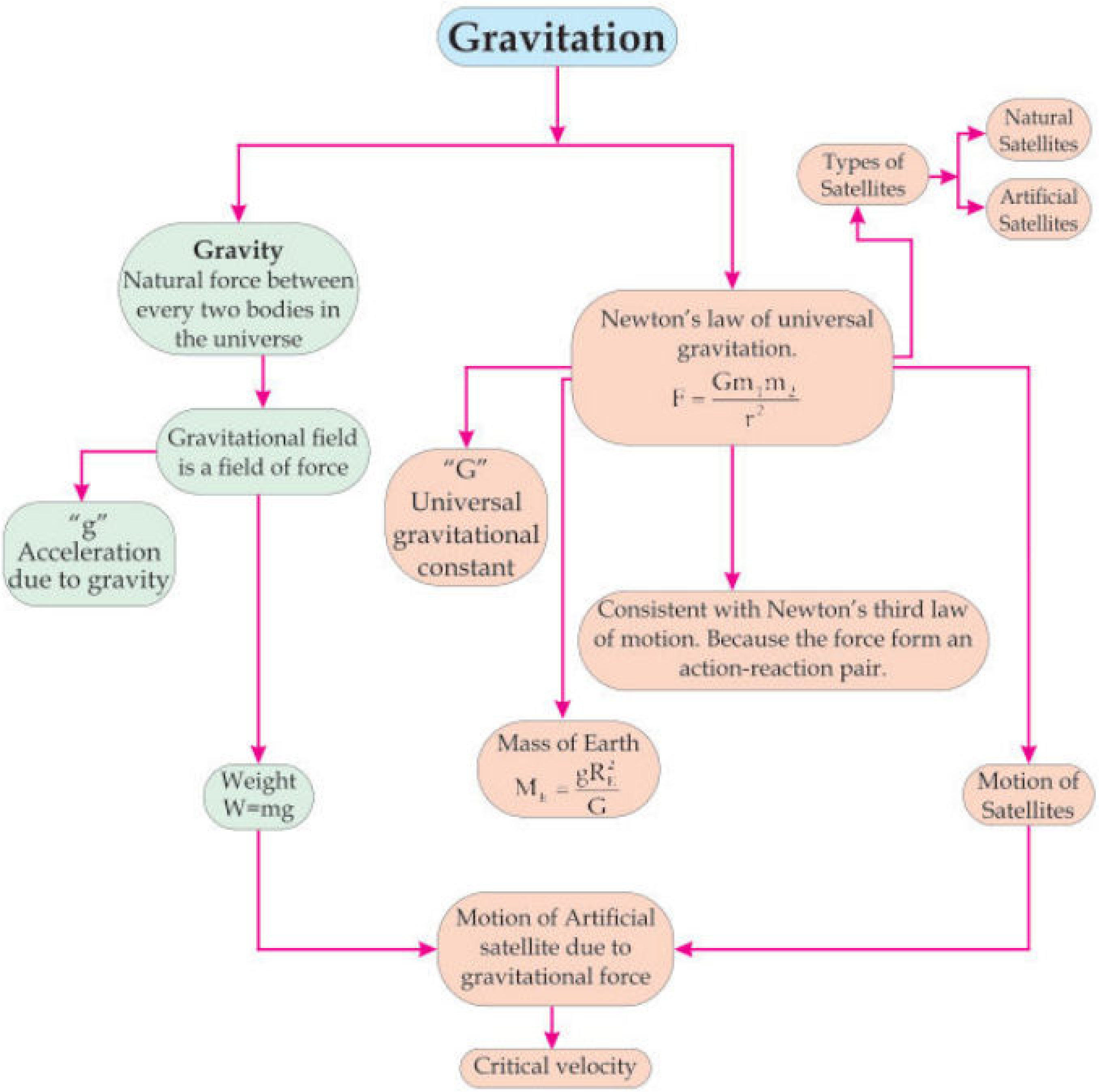
SUMMARY

- The gravitational force (pull) of Earth is known as gravity.
- Everybody in the universe attracts every other body with a gravitational force of magnitude.

$$F = \frac{Gm_1m_2}{r^2}$$

- Gravitational force forms an action-reaction pair. Newton's law of gravitation is consistent with Newton's third law of motion.
- "G" has constant value through out the universe.
- "g" has different values at different places.
- A gravitational field is a region in which a mass is attracted due to gravitational attraction.
- Weight of an object is the gravitational pull of Earth acting on it. Mathematically, $W = mg$.
- Mass of Earth is 6.0×10^{24}kg.
- A satellite is an object that revolves around a planet.
- Natural satellite is a planet that revolves around another planet naturally, like Moon is natural satellite of Earth.
- An artificial satellite is an object which is sent to space to revolve around a planet, like Sputnik-1, Meteosat are artificial satellites of Earth.
- Critical velocity is the constant horizontal velocity needed to put a satellite into an stable circular orbit around the Earth.

CONCEPT MAP


Gravitation
Gravity
Natural force between every two bodies in the universe
Gravitational field is a field of force
"g" Acceleration due to gravity
Weight W=mg
Newton's law of universal gravitation. $F = \frac{Gm_1m_2}{r^2}$
"G" Universal gravitational constant
Consistent with Newton's third law of motion. Because the force form an action-reaction pair.
Mass of Earth $M_E = \frac{gR_E^2}{G}$
Types of Satellites
Natural Satellites
Artificial Satellites
Motion of Satellites
Motion of Artificial satellite due to gravitational force
Critical velocity

End of Unit Questions

Section (A) Multiple Choice Questions (MCQs)

1. The motion of a falling ball towards Earth is due to the -----------.
   a) Weightlessness
   b) Gravitational force
   c) Acceleration due to gravity
   d) Both 'a' and 'b'
2. Newton's law of gravitation holds between every two objects on the -----------.
   a) on Earth b) on Jupiter
   c) on Moon d) on Universe
3. Numerical value of G is -------------.
   a) $G = 6.673 \times 10^{\ 11} Nm^2\ kg^{\ 2}$
   b) $G = 6.673 \times 10^{11} Nm^2\ kg^{\ 2}$
   c) $G = 6.763 \times 10^{-10} Nm^2\ kg^{-2}$
   d) $G = 6.763 \times 10^{11} Nm^2\ kg^{-2}$
4. Gravitational field of Earth is directed --------------.
   a) towards the Earth b) towards the Sun
   c) towards the Moon d) away from Earth
5. -------------------- was the first scientist who gave the concept of gravitation.
   a) Einstein b) Newton
   c) Faraday d) Maxwell
6. According to Newton's law of universal gravitation force ($\propto$) _______.
   a) $m_1 m_2$ b) $\frac{1}{r^2}$
   c) r^2 d) Both (a) and (b)
7. Gravitational force is always -------------.
   a) Repulsive b) Attractive
   c) Both a and b d) None of these

8. Numerical value of ----------------- remains constant every where.
   a) g b) G
   c) F d) W
9. Gravitation force is ------------- of the medium between the objects.
   a) Dependent b) Independent
   c) Both 'a' and 'b' d) None of these
10. Near Earth's surface (g) = -------------.
   a) $10ms^{-2}$ b) $1.6ms^{-1}$
   c) Both (a) and (b) d) None of these
11. Newton's law of gravitation is consistent with Newton's ------------------ law of motion.
   a) 1^{st} b) 2^{nd}
   c) 3^{rd} d) All of them
12. Spring balance is used to measure ---------------.
   a) Mass b) Weight
   c) Elasticity d) Density
13. Your weight as measured on Earth will be --------- on Moon.
   a) Increased b) Decreased
   c) Remains same d) None of these
14. Mass of Earth is ----------------.
   a) 6.0×10^{23}kg b) 6.0×10^{24}kg
   c) 6.0×10^{25}kg c) 6.0×10^{26}kg
15. ------------- is a natural satellite.
   a) Earth b) Jupiter
   c) Moon d) Mars
16. A communication satellite completes its one revolution around the Earth in --------------- hours.
   a) 6 b) 12
   c) 18 d) 24
17. The velocity of a satellite is ------------- of its mass.
   a) Independent b) Dependent
   c) Equal d) Double

18. ----------------- are used to put satellites into orbits.
   a) Helicopter b) Aeroplane
   c) Rocket d) None of these
19. The critical velocity (v_c) = --------------.
   a) gR b) $\frac{g}{R}$
   c) $\sqrt{gR}$ d) $\sqrt{\frac{g}{R}}$

Section (B) Structured Questions

Newton's Law of Gravitation

1. a) Why we do not feel the gravitational force of attraction from the objects around us?
   b) Define Gravitational field with an example.
2. a) Write down any three characteristics of Gravitational force.
   b) Define gravitational field strength.
3. a) State and explain Newton's law of gravitation.
   b) Define 'field force'.
4. Determine the gravitational force of attraction between Urwa and Ayesha standing at a distance of 50m apart. The mass of Urwa is 60kg and that of Ayesha is 70kg.

Weight

5. a) Why weight of an object does not remain same every where on Earth?
   b) Why the unit of weight is Newton? Explain.
6. a) Define weight and write down its equation.
   b) Weight of Rani is 450N at the surface of Earth. Find her mass?
7. Weight of Naveera is 700N on the Earth's surface. What will be Naveera's weight at the surface of Moon?

8. a) Your weight decreases as you go up at high altitudes, without dieting. Explain.
   b) If you step on a scale and it gives reading 55kg, is that a measure of your weight. If not then which physical quantity it shows?

Mass of Earth

9. Calculate the mass of Earth by using Newton's law of gravitation.
10. If "M_E" is the mass of Earth, "R_E" radius of Earth, "G" is universal gravitational constant, then find acceleration due to gravity "g";
    i) On the surface of Earth.
    ii) At the centre of Earth.
11. A planet has mass four times of Earth and radius two times that of Earth. If the value of "g" on the surface of Earth is $10ms^{-2}$. Calculate acceleration due to gravity on the planet.
12. Evaluate the acceleration due to gravity in terms of mass of Earth "M_E", radius of Earth "R_E" and universal gravitational constant "G":
    i) At a distance, twice the Earth's radius.
    ii) At a distance, one half the Earth's radius.

Artificial Satellite

13. a) Calculate the speed of a satellite which orbits the Earth at an altitude of 400 kilometers above Earth's surface.
    b) Write the name of any one natural satellite.
14. a) Write down the names of four different types of orbit.
    b) Define the terms
       i) Critical Velocity.
       ii) Communication Satellite.

15. Derive the expression for the motion of a satellite.

$$v = \sqrt{\frac{GM}{R + h}}$$

16. a) Differentiate between the natural satellite and artificial satellite.
    b) Name the parameters on the basis of which orbits are characterized?

Unit - 7

PROPERTIES OF MATTER

Matter is made up of tiny particles called atoms.

Matter exists in different states. three basic states of matter are solid liquid and gas.

The properties of matter in these states can be described on the basis of the forces and distances between their molecules and energy of the molecules..

The temperature and pressure of a gas depends upon the motion of its molecules.

A change in volume of a fixed mass of a gas at constant temperature is caused by a change in pressure applied to the gas.

This fact is used in many fields of daily life. For example, in using syringe, in pumping air to the tyre through a bicycle pump, in spraying color etc.

Matter can change its state and water is the best example of it.

Students Learning Outcomes (SLOs)

After learning this unit students should be able to:

- Describe States of matter.
- State kinetic molecular model of matter.
- Explain the kinetic model in terms of forces between particles.
- Explain the behavior of gases.
- Calculate changes in pressure and volume:

$$P_1 V_1 = P_2 V_2$$